GSW Vascular Injury

Section 1: Case Summary

Scenario Title:	Gunshot wound with vascular injury
Keywords:	Gunshot wound, GSW, peripheral vascular injury
Brief Description of Case:	Adult male with penetrating chest and extremity trauma (GSW) with vascular compromise, requiring a thorough and systematic approach. For senior learners, there will be an additional element of instability from the penetrating chest injury requiring chest tube.

Goals and Objectives	
Educational Goal:	Systematic ATLS approach to trauma Approach to penetrating peripheral extremity trauma Approach to penetrating chest trauma *Optional considerations for difficulty if senior learner:* - Community hospital with transport considerations - Adding additional injuries - Sim actor -> resident who does not know ATLS or inexperienced RN
Objectives: (Medical and CRM)	Medical objectives: 1. Demonstrate a systematic ATLS approach in a case of penetrating trauma. 2. Assess and manage penetrating chest trauma according to major trauma guideline algorithms (e.g., ATLS, EAST or WEST) 3. Assess and manage penetrating extremity trauma and peripheral vascular injury according to major trauma guideline algorithms (e.g., ATLS, EAST or WEST) CRM objectives: 1. Task prioritizing – systematic approach to trauma with prioritizing of life-threatening findings 2. Situational awareness – maintain situational awareness while completing a systematic approach to trauma 3. Communication – use concise and effective critical language 4. Followership – use a clear and loud voice during ATLS assessment
EPAs Assessed:	F1: Initiating and assisting in resuscitation of critically ill patients C2. Resuscitating and coordinating care for critically injured trauma patients C5: Identifying and managing patients with emergent medical or surgical conditions C13: Performing advanced procedures

Learners, Setting and Personnel				
Target Learners:	☒ Junior Learners	☒ Senior Learners	☐ Staff	
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:			
Location:	☒ Sim Lab	☐ In Situ	☐ Other:	
	Instructors: 1			

Recommended Number of Facilitators:	Sim Actors: 1
	Sim Techs: 1

Scenario Development	
Date of Development:	2021-01-16
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Last Revision Date:	-
Revised By:	-
Version Number:	1

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Alex			Age: 26	Gender: M	Weight: 210 lbs
Presenting complaint: Gunshot wounds					
Temp: 36.8	HR: 115	BP: 110/65	RR: 26	O_2Sat: 95%	FiO_2: 21% (RA)
Cap glucose: 5.0 mmol/L			GCS: 14 (E4 V4 M6)		
Triage note: Multiple gunshot wounds. “Minding my own business” downtown. Dropped off by car at emergency doors, drop and run. Patient intoxicated.					
Allergies: Unknown					
Past Medical History: Unknown			Current Medications: Unknown		

Section 2B: Extra Patient Information

A. Further History
Patient is intoxicated, agitated, loud, and difficult but not violent. He will answer direct questions, but he is very evasive regarding what actually happened, saying he was “minding my own business when someone came out from nowhere and shot me”. The weapon was a handgun at about 1-2m away. He thinks the gun was fired about “5 or 6 times, but I’m not sure, it was all so fast”. He will not answer any questions about who dropped him off at the ED. Complaining of leg pain. He refuses attempts to examine it unless analgesia is given and he is prompted multiple times. If directly asked, he drank about “3 or 4 shots” at a bar tonight. He also smoked cannabis. No other drug use.

B. Physical Exam	
Cardio: Tachycardia	Neuro: No findings
Resp: Tachypnea, GSW as described below	Head & Neck: No findings
Abdo: No findings	MSK/skin: *see below for description of GSWs*
Other: Gunshot wounds x3 - Left leg GSW (anterior mid-thigh) that is actively bleeding - Left leg GSW (posterior upper-thigh/buttock) that is actively bleeding. Only visible if leg is raised or patient is rolled. - Left clavicle / upper torso GSW with bleeding controlled. Breath sounds initially present bilaterally. The distal left leg cool, but has a pulse, intact neurologic function, and he does not have pulsatile bleeding or a rapidly expanding hematoma. POCUS: Absent lung sliding on left. No pericardial effusion. No free fluid in abdomen.	

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin - adult
☒ Standardized Patient – *either mannequin or standardized patient, depending on degree of intended stress on team's situational awareness.*
☐ Task Trainer
☒ Hybrid – *optional hybrid task trainer for chest tube*

B. Special Equipment Required
Handheld doppler ultrasound for pulse check Ultrasound for POCUS 14-16g needle for needle decompression Chest tube tray Tourniquet

C. Required Medications
Crystalloid fluids Blood Fentanyl, Cephazolin, TXA

D. Moulage
Moulage for gunshot wounds: - Left leg wound (front thigh) – bleeding; large volume of blood on sheets/leg, sheet underneath is soaked - Left lower back/buttock wound (back) – bleeding. Should only be visible if the leg is lifted or patient rolled. - Left clavicle region (front) Patient should be wearing clothing that needs to be removed.

E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor

F. Patient Reactions and Exam
Patient responses: Complains of leg pain. Patient is intoxicated, agitated, and loud. He will answer direct questions, but he is very evasive. He says he was "minding my own business when someone came out from nowhere and shot me". He will not answer any questions about who dropped him off at the ED. **Exam:** HR 110, RR 26 Left leg gunshot wound anterior mid-thigh that is actively bleeding Left leg gunshot wound posterior upper-thigh/buttock that is actively bleeding Left clavicle / upper torso gunshot wound with bleeding controlled, and breath sounds present bilaterally. Distal left leg is cool but has a pulse and he can move it.

Section 4: Sim Actors and Standardized Patients

Sim Actors and Standardized Patient Roles and Scripts	
Role	*Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)*
Nurse	Assists with standard resuscitation equipment, medications, and adds to fidelity in terms of patient's level of agitation and cooperation. If using a mannequin (rather than an actor), the controller should be very vocal to maintain fidelity. The sim actor should help establish the difficulty managing the patient (e.g., "Sir, please lay back so we can take care of you! You have to let us assess you or we can't help!") May need to provide cues regarding physical exam findings: - Gunshot wounds x3 o Left leg GSW (anterior mid-thigh) that is actively bleeding o Left leg GSW (posterior upper-thigh/buttock) that is actively bleeding o Left clavicle / upper torso GSW with bleeding controlled - Left leg is cool, but has a pulse and intact neurologic function. He does not have pulsatile bleeding or a rapidly expanding hematoma. o ABI for left leg is 0.7. o If team does not ask for ABI after initial management, RN should prompt "I've seen TTLs ask for ABIs for this, should we do one?"
Patient (if using mannequin)	If using a mannequin (rather than an actor), the controller should be very vocal to maintain fidelity.
Patient (if using SP)	Patient is intoxicated, agitated, loud, and difficult but not violent. He will answer direct questions. Complaining of leg pain, and extreme pain if it is touched. Refusing attempts to examine it, but eventually will allow it if given pain medications and talked to about it multiple times. The left leg should be the patient's primary concern and he will redirect attention to it constantly. In stage 2 – the patient becomes short of breath due to a developing tension pneumothorax.

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus tach HR: 110 BP: 110/65 RR: 26 O_2SAT: 96% T: 36.8^{o}C GCS: 14	Intoxicated Left leg pain Actively bleeding from left leg	Expected Learner Actions ☐ Gather history from patient ☐ ATLS assessment (i.e. ABCDE) ☐ Monitors, vitals, IV access ☐ Trauma code ☐ Calling for blood ☐ Bloodwork - trauma panel ☐ Cefazolin ☐ TXA ☐ Analgesia (e.g., fentanyl) ☐ POCUS eFAST (absent left lung sliding) ☐ Call for Xray (CXR, pelvis, leg) ☐ Order CT chest and CT-A of left leg	Modifiers - Screams in pain and refuses assessment if team touches left leg - BP slowly declines 10mmHg every 2 min until pressure applied to BOTH front and back of left leg Triggers - once analgesia given, patient will allow assessment of left leg - full exposure completed and pressure applied to left leg → go to stage 2	Residents should recognize initial vitals as indication to call for blood He has ongoing bleeding from his leg, which will be controlled with compression or any hemostatic intervention. Could consider applying paper clip to map bullet trajectory on CXR
2. Unstable HR: 125 BP: 85/55 RR: 30 O_2SAT: 90%	Unstable vitals Loses breath sounds on left	Expected Learner Actions ☐ Oxygen applied ☐ Massive transfusion protocol ☐ Left chest tube +/- needle or finger thoracotomy	Modifiers - If team calls for consults, they are up in the OR - After blood and decompress chest → HR 110, BP 100/55, sat 96% Triggers - Ask for CT-A, or if 4min pass → go to stage 3 (leg pulseless)	Xrays are given to team

3. Pulseless leg Rhythm: Sinus tach HR: 110 BP: 100/55 RR: 26 O_2SAT: 96%	Left leg now more mottled, cold, pulseless and pain worse	Expected Learner Actions ☐ Recheck leg pulse (absent) ☐ Doppler U/S to confirm absence of pulse ☐ Consult Orthopedic / Vascular surgery emergently	Modifiers - If team uses Doppler, no pulse is found. Triggers - Team asks for consults emergently → Trauma team and orthopedics arrive to take patient to OR	- RN should prompt that leg is now appearing mottled / pale / cold

Appendix A: Laboratory Results

<u>CBC</u> WBC: **14** Hgb: 120 Plt: 276 <u>Lytes</u> Na 135 K 4.5 Cl 109 HCO_3 23 AG 3 Urea 7 Cr 80 Glucose 5.5 <u>VBG</u> pH **7.34** pCO_2 30 pO_2 80 HCO_3 20 Lactate **2.0**	<u>Biliary</u> AST **72** ALT 35 GGT **120** ALP 40 Bili 8 Lipase 20 <u>Tox</u> EtOH **35** ASA <1 Tylenol <1 Dig level <1 Osmols 290

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

CXR for juniors – Left clavicle fracture + bullet

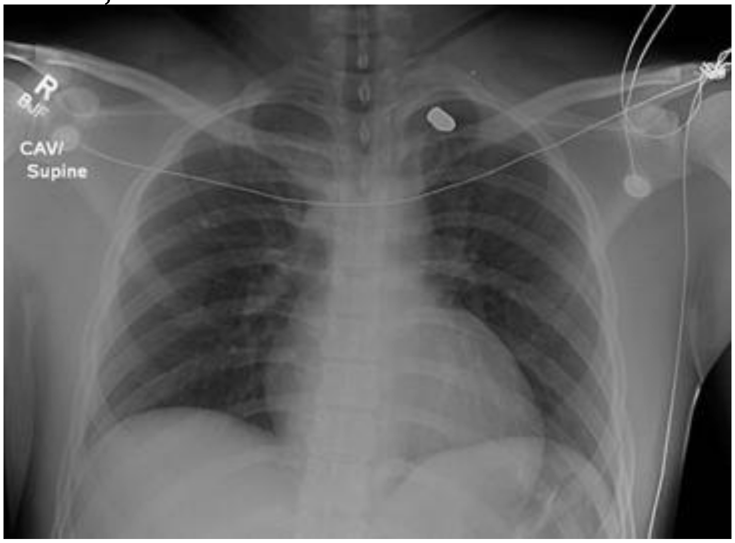


Image from: https://www.sciencedirect.com/science/article/pii/S1930043315305197

CXR for seniors – hemopneumothorax + bullet

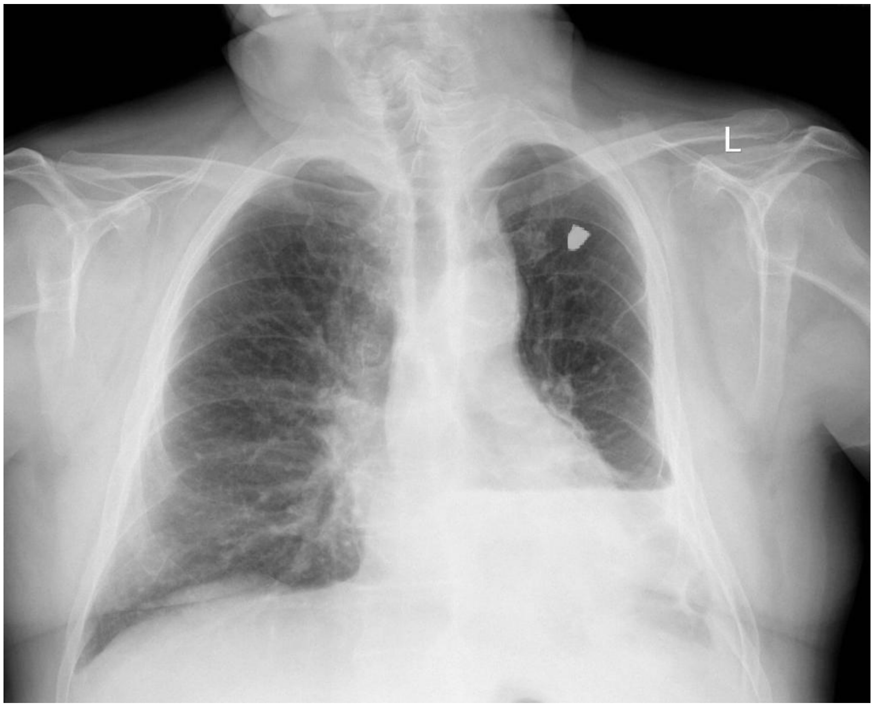


Image from: http://learningradiology.com/archives2007/COW%20265-Hemothorax/hemothoraxcorrect.html

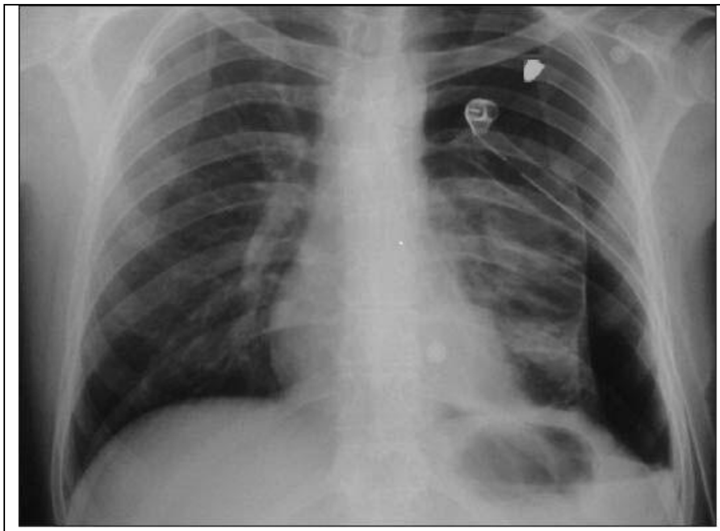


Image from: https://www.researchgate.net/figure/chest-X-ray-shows-persistent-pneumothorax-after-chest-tube-insertion_fig4_260486588

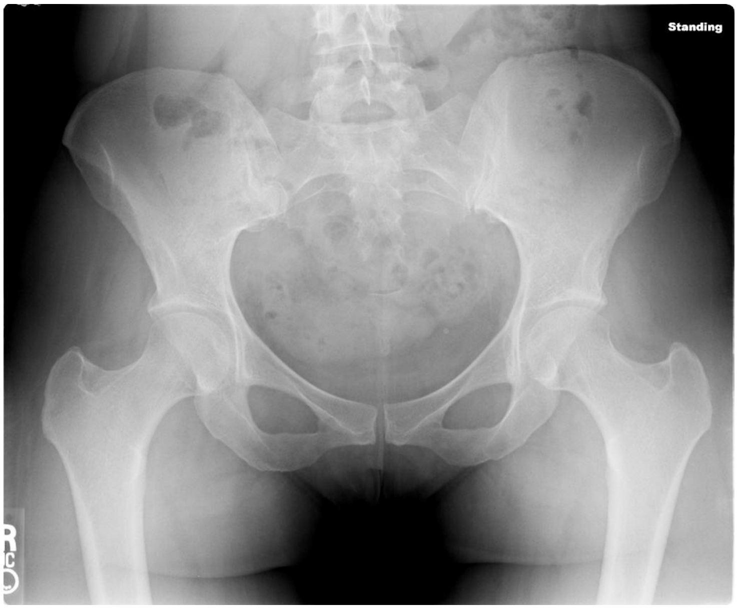


Image from: https://radiopaedia.org/cases/normal-pelvis-x-ray-ap

Image from: https://www.orthobullets.com/trauma/1059/gun-shot-wounds

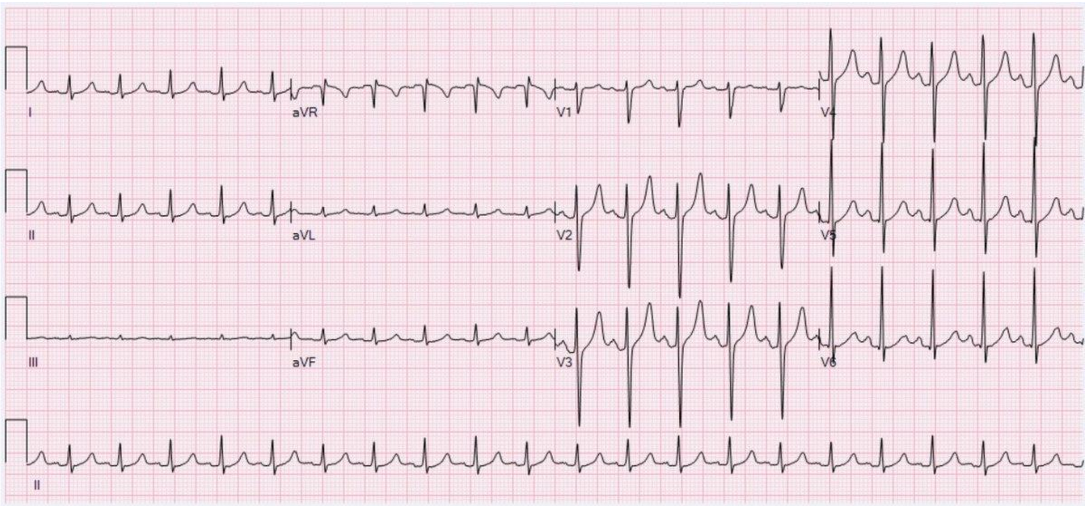


Image from: https://en.wikipedia.org/wiki/Sinus_tachycardia

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

CRM objectives:

1. Task prioritizing – discuss a systematic approach to trauma with prioritizing of life-threatening findings (i.e. ATLS approach). Learners may fail to recognize source of bleeding on posterior leg if they do not fully examine the leg or log roll early enough (importance of good exposure).
2. Situational awareness – this may be challenged by multiple concurrent findings and the patient's behaviour.
3. Communication and Followership – use a clear and loud voice during ATLS assessment, and critical language when important status changes are noted (e.g. leg becoming pulseless)

Medical objectives:

1. Be able to assess and management penetrating extremity trauma and peripheral vascular injury according to major trauma guideline algorithms (e.g., ATLS, EAST or WEST)
2. Be able to assess and manage penetrating chest trauma according to major trauma guideline algorithms (e.g., ATLS, EAST or WEST)

Key Points from EAST (2012) document on Penetrating Lower Extremity Arterial Trauma:
Level 1 recommendations:

- CTA may be used as the primary diagnostic study for evaluation of penetrating lower extremity vascular injury when imaging is required.

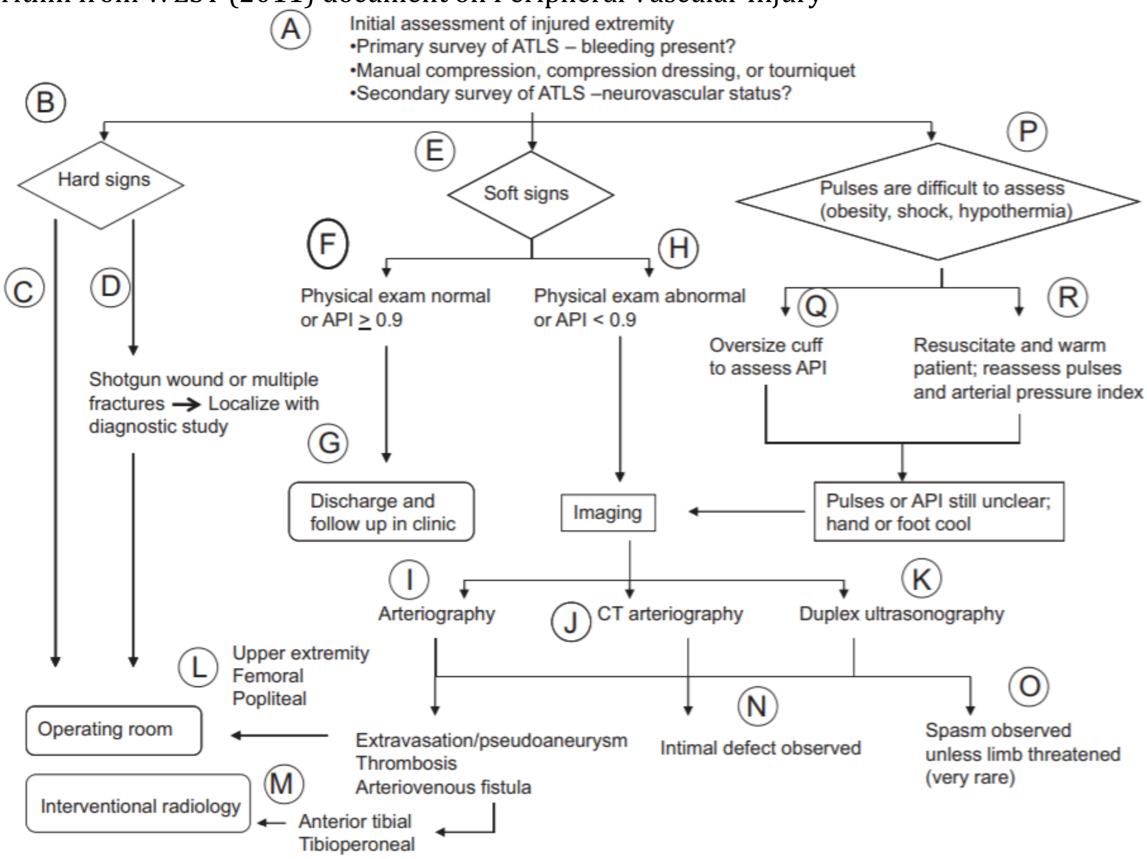
Level 2 recommendations:

- With hard signs --> should be surgically explored. There is no need for arteriogram in this setting unless the patient has an associated skeletal or shotgun injury. Restoration of perfusion to an extremity with an arterial injury should be performed in less than 6 hours to maximize limb salvage (2002).
- Without hard signs --> those who have abnormal physical examination findings and/or an ABI < 0.9 should have further evaluation to rule out vascular injury.
- Without hard signs, with normal exam and ABI >0.9 --> may be discharged

Level 3 recommendations:

- In cases of hemorrhage from penetrating lower extremity trauma in which manual compression is unsuccessful, tourniquets may be used as a temporary adjunct for hemorrhage control until definitive repair.
- The role of non-invasive Doppler pressure monitoring or duplex ultrasonography to confirm or exclude arterial injury is not well defined. There may be a role for these studies in patients with soft signs of vascular injury or with proximity injuries (2002).
- Nonoperative observation of asymptomatic nonocclusive arterial injuries is acceptable (2002).
- Arteriography for proximity is indicated only in patients with shotgun injuries (2002).

Key algorithm from WEST (2011) document on Peripheral Vascular Injury



Algorithm for evaluation of patient with possible peripheral vascular injury.

Box 8
Hard and soft signs of extremity arterial injury

Hard Signs	Soft Signs
Absent distal pulse	History of significant blood loss at the scene
Active/pulsatile bleeding	Proximity of a penetrating wound or blunt trauma to an artery
Rapidly expanding hematoma	Nonpulsatile hematoma
Classic signs of distal ischemia, 5Ps: pallor, pain, paresthesias, paralysis, pulseless	Neurologic deficit attributed to a nerve adjacent to a named artery
Palpable thrill or audible bruit over the injured area	Delayed capillary refill
	Diminished pulse compared with contralateral extremity

Data from Callcut RA, Mell MW. Modern advances in vascular trauma. Surg Clin North Am 2013;93(4):941–61, ix.

Clinical Features of Vascular Injury

HARD FINDINGS	SOFT FINDINGS
Pulsatile hemorrhage	History of significant hemorrhage at scene
Expanding hematoma	Nonexpanding hematoma
Absent distal pulses	Diminished pulse or ABI of injured extremity
Palpable thrill	Extremity peripheral nerve deficit
Audible bruit	Bony injury or proximate penetrating wound

ABI, Ankle-brachial index.

Key algorithm from WEST (n.d., retrieved 2020)

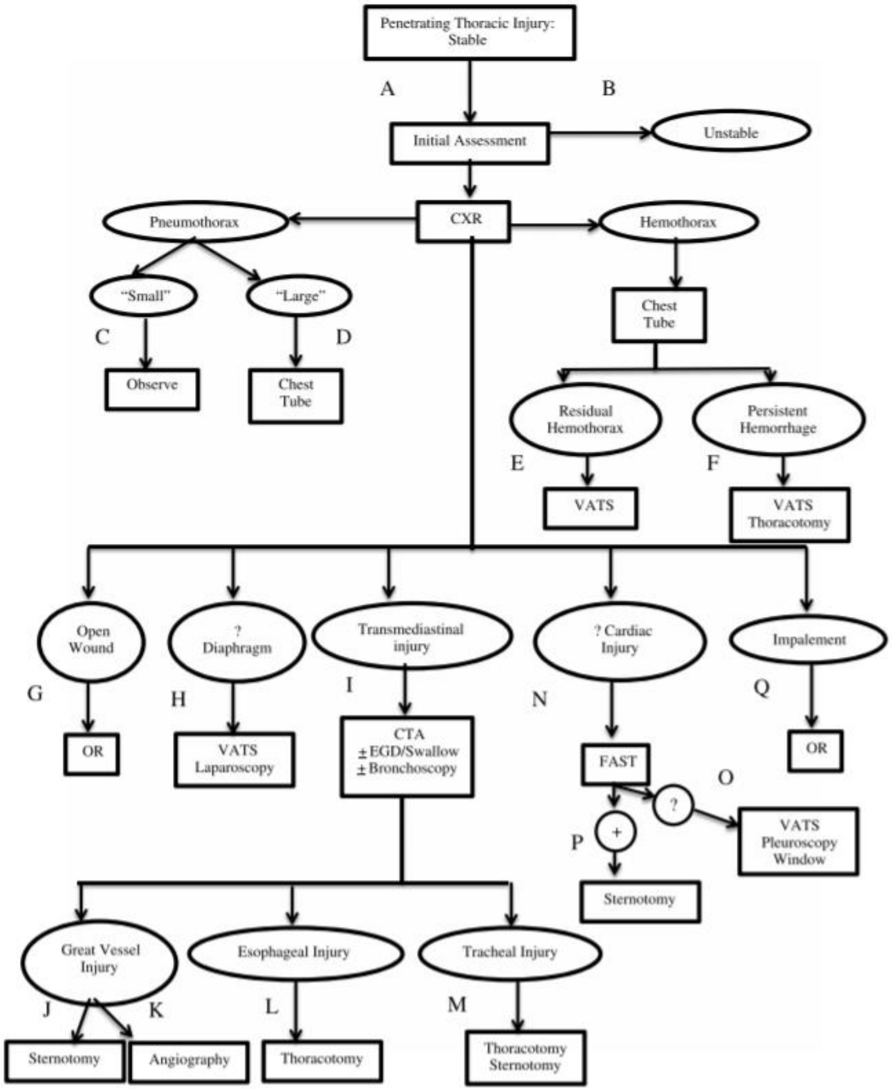


Figure 2. Management of the unstable patient.

Exam

- inspect widely for any other penetrating wounds (e.g. entry/exit, second wounds, fragments), including looking at back, buttocks, and perineum
- palpating the penetrating wound is not helpful and is very painful
- mark wounds with radio-opaque marker (e.g. paper clip, staple), to assist radiographically
- number of holes + number of missiles should yield an even number

"Danger zone" or "cardiac box" has been described as the region between the epigastrium to the sternal notch and laterally within 3 cm of the sternum

References

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